

SQL Code

```
1  CREATE DATABASE TechNovaDB;
2  USE TechNovaDB;
3
4  CREATE TABLE Department (
5      DeptID INT PRIMARY KEY,
6      DeptName VARCHAR(50) NOT NULL,
7      Location VARCHAR(50) NOT NULL
8  );
9
10 CREATE TABLE Employee (
11     EmpID INT PRIMARY KEY,
12     EmpName VARCHAR(100) NOT NULL,
13     Gender CHAR(1) CHECK (Gender IN ('M', 'F', 'O')),
14     DOB DATE NOT NULL,
15     HireDate DATE NOT NULL,
16     DeptID INT,
17     FOREIGN KEY (DeptID) REFERENCES Department(DeptID) ON DELETE SET NULL
18 );
19
20 CREATE TABLE Project (
21     ProjectID INT PRIMARY KEY,
22     ProjectName VARCHAR(100) NOT NULL,
23     DeptID INT,
24     StartDate DATE NOT NULL,
25     EndDate DATE,
26     FOREIGN KEY (DeptID) REFERENCES Department(DeptID) ON DELETE CASCADE
27 );
28
29 CREATE TABLE Performance (
30     EmpID INT,
31     ProjectID INT,
32     Rating DECIMAL(3,2) CHECK (Rating BETWEEN 1.00 AND 5.00),
33     ReviewDate DATE NOT NULL,
34     PRIMARY KEY (EmpID, ProjectID),
35     FOREIGN KEY (EmpID) REFERENCES Employee(EmpID) ON DELETE CASCADE,
36     FOREIGN KEY (ProjectID) REFERENCES Project(ProjectID) ON DELETE CASCADE
37 );
38
39 CREATE TABLE Reward (
40     RewardID INT AUTO_INCREMENT PRIMARY KEY,
41     EmpID INT,
42     RewardMonth VARCHAR(20) NOT NULL,
43     RewardAmount DECIMAL(10,2) NOT NULL,
44     RewardYear INT NOT NULL,
45     FOREIGN KEY (EmpID) REFERENCES Employee(EmpID) ON DELETE CASCADE
46 );
47
48 CREATE INDEX idx_emp_name ON Employee(EmpName);
49 CREATE INDEX idx_emp_dept ON Employee(DeptID);
50
51 INSERT INTO Department (DeptID, DeptName, Location) VALUES
52 (101, 'IT', 'Bangalore'),
53 (102, 'HR', 'Delhi'),
54 (103, 'Finance', 'Mumbai'),
55 (104, 'Marketing', 'Hyderabad'),
56 (105, 'Sales', 'Pune');
57
58 INSERT INTO Employee (EmpID, EmpName, Gender, DOB, HireDate, DeptID) VALUES
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59 (1, 'Abhishek', 'M', '2003-01-18', '2026-01-10', 101),
60 (2, 'Rashi', 'F', '1988-04-09', '2025-03-22', 102),
61 (3, 'Nisha', 'F', '1995-01-15', '2022-08-05', 101),
62 (4, 'Vikas', 'M', '1992-11-23', '2018-04-12', 103),
63 (5, 'Ashok', 'M', '1994-05-19', '2026-01-15', 104);
64
65 INSERT INTO Project (ProjectID, ProjectName, DeptID, StartDate, EndDate) VALUES
66 (201, 'Cloud Migration', 101, '2025-01-10', '2025-08-20'),
67 (202, 'Talent Acquisition Plan', 102, '2025-02-01', '2025-06-01'),
68 (203, 'Automation Tool', 101, '2025-03-15', NULL),
69 (204, 'Annual Audit 2026', 103, '2026-01-05', '2026-03-30'),
70 (205, 'Digital Campaign', 104, '2025-09-01', '2025-12-31');
71
72 INSERT INTO Performance (EmpID, ProjectID, Rating, ReviewDate) VALUES
73 (1, 201, 4.50, '2025-08-22'),
74 (2, 202, 3.80, '2025-06-05'),
75 (3, 203, 4.80, '2025-12-20'),
76 (4, 204, 4.20, '2026-04-02'),
77 (5, 205, 3.50, '2026-01-05');
78
79 INSERT INTO Reward (EmpID, RewardMonth, RewardAmount, RewardYear) VALUES
80 (1, 'March', 2500.00, 2026),
81 (3, 'April', 3000.00, 2026),
82 (4, 'January', 800.00, 2025),
83 (2, 'February', 1500.00, 2026);
84
85 UPDATE Employee SET DeptID = 104 WHERE EmpID = 2;
86 DELETE FROM Reward WHERE RewardAmount < 1000.00;
87 -- SET SQL_SAFE_UPDATES = 0; to enable put value 1;
88
89
90
91
92 -- all employees who joined after 2019-01-01
93 SELECT * FROM Employee
94 WHERE HireDate > '2019-01-01';
95
96 -- average performance rating of employees in each department
97 SELECT e.DeptID, d.DeptName, AVG(p.Rating) AS AverageRating
98 FROM Performance p
99 JOIN Employee e ON p.EmpID = e.EmpID
100 JOIN Department d ON e.DeptID = d.DeptID
101 GROUP BY e.DeptID, d.DeptName;
102
103 -- List employees with their calculated age
104 SELECT EmpID, EmpName, DOB,
105 TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS Age
106 FROM Employee;
107
108 -- total rewards given in the current year (2026)
109 SELECT YEAR(CURDATE()) AS CurrentYear, SUM(RewardAmount) AS TotalRewards
110 FROM Reward
111 WHERE RewardYear = YEAR(CURDATE());
112
113 -- Retrieve employees who have received rewards greater than 2000
114 SELECT DISTINCT e.EmpID, e.EmpName, r.RewardAmount
115 FROM Employee e
116 JOIN Reward r ON e.EmpID = r.EmpID
117 WHERE r.RewardAmount > 2000.00;
118
119
120

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121
122 -- Display Employee Name, Department Name, Project Name, and Rating
123 SELECT e.EmpName, d.DeptName, pr.ProjectName, p.Rating
124 FROM Employee e
125 JOIN Department d ON e.DeptID = d.DeptID
126 JOIN Performance p ON e.EmpID = p.EmpID
127 JOIN Project pr ON p.ProjectID = pr.ProjectID;
128
129 -- the highest-rated employee in each department using a subquery
130 SELECT e.EmpName, d.DeptName, p.Rating
131 FROM Employee e
132 JOIN Department d ON e.DeptID = d.DeptID
133 JOIN Performance p ON e.EmpID = p.EmpID
134 WHERE p.Rating = (
135     SELECT MAX(p2.Rating)
136     FROM Performance p2
137     JOIN Employee e2 ON p2.EmpID = e2.EmpID
138     WHERE e2.DeptID = e.DeptID
139 );
140
141 -- List all employees who have not received any rewards using a subquery
142 SELECT EmpID, EmpName FROM Employee
143 WHERE EmpID NOT IN (SELECT DISTINCT EmpID FROM Reward);
144
145
146 ----- transaction-----
147
148
149 START TRANSACTION;
150 INSERT INTO Employee (EmpID, EmpName, Gender, DOB, HireDate, DeptID) VALUES (6, 'Siddharth', 'M', '1993-08-
151 14', CURDATE(), 101);
152 INSERT INTO Performance (EmpID, ProjectID, Rating, ReviewDate) VALUES (6, 201, 4.10, CURDATE());
153 COMMIT;
154
155 EXPLAIN SELECT e.EmpName, d.DeptName, p.Rating FROM Employee e FORCE INDEX (PRIMARY)
156 JOIN Department d ON e.DeptID = d.DeptID
157 JOIN Performance p ON e.EmpID = p.EmpID
158 WHERE e.DeptID = 101;
159
160 EXPLAIN SELECT e.EmpName, d.DeptName, p.Rating FROM Employee e USE INDEX (idx_emp_dept)
161 JOIN Department d ON e.DeptID = d.DeptID
162 JOIN Performance p ON e.EmpID = p.EmpID
163 WHERE e.DeptID = 101;
164
165 SELECT * FROM Employee ;
166 SELECT * FROM Performance;

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